

University of Calgary

Faculty of Computer Science

Final report for CPSC601: Connectomics

Instructor: Professor Emma Towlson

Comparative Network Analysis of Drosophila Larval and Adult Connectomes

by Parham Moonesisohi

December 6, 2025

ABSTRACT

Study of segregation and integration and their correlation with different cognitive states in human has been a popular trend in connectomics research. However, due to lack of enough structural connectomes, these studies never worked on structural data. On the other hand, in the past few years, many detailed structural connectomes have been released, and two of those are larvae and adult *Drosophila* connectomes. This allowed us to study the differences between the adult and larvae, with integration and segregation in mind. To do that, we use:

- null models that specifically rewire different types of edges, helping us to understand the underlying cause of the significance, if there's any.
- SNAP library and ARC to compute the statistics for the adult connectome and its null models due to its enormous size.
- measures that have been used to study integration and segregation in human functional connectomes; such as efficiency, centrality and rich-clubs.

Our research questions are:

- Is the adult brain more segregated than the larvae?
- Do local connections have a different effect on segregation, than the inter-modular connections?
- Does the adult brain develop more connector hubs in order to facilitate integration?

Contents

1	Introduction	1
1.1	Integration and Segregation	1
1.2	Drosophila Melanogaster	2
2	Data Description	3
2.1	Larvae	3
2.2	Adult	3
3	Basic Stats	5
3.1	Degree Distribution	5
3.2	Stats	6
4	Methods	7
4.1	Consensus Clustering	7
4.2	Null Models	8
4.3	Measures	9
4.4	Parallelization	10
5	Visualization	11
6	Results	13
6.1	Clustering Coefficient and Local Efficiency	13
6.2	Global Efficiency	14
6.3	Centrality Measures	14
6.3.1	Betweenness Centrality	14
6.3.2	Eigenvector Centrality	16
6.3.3	Closeness Centrality	17
6.4	Rich-Club	17
7	Conclusions	19
	Bibliography	21
	Code Availability	23

List of Figures

3.1	Larvae degree distribution. Axes are in log scale.	5
3.2	Adult degree distribution. Axes are in log scale.	6
4.1	Diagram explaining the flow of the consensus clustering algorithm. .	8
5.1	Gephi visualization for the larvae connectome. Only the top 13.5% of the nodes with the highest degree are visible. Colors show the consensus communities.	11
5.2	Gephi visualization for the adult connectome. Only the top 0.5% of the nodes with the highest degree are visible. Colors show the consensus communities.	12
6.1	Box plot comparing the original average clustering coefficient of both connectomes with their null models.	13
6.2	Box plot comparing the original average local efficiency of both connectomes with their null models.	14
6.3	Scatter plot showing the correlation between the local efficiency and the clustering coefficient in both connectomes.	14
6.4	Box plot comparing the original global efficiency of both connec- tomes with their null models.	15
6.5	Mean betweenness of nodes in a network. The box plot is comparing both connectomes with their null models.	16
6.6	Maximum betweenness of nodes in a network. The box plot is comparing both connectomes with their null models.	16
6.7	Mean betweenness of nodes in a network. The box plot is comparing both connectomes with their null models.	17
6.8	Mean closeness of nodes in a network. The box plot is comparing both connectomes with their null models.	18
6.9	Mean closeness of nodes in a network. The box plot is comparing both connectomes with their null models.	18

List of Tables

2.1	Summary of Datasets	4
3.1	Basic Statistics	6
6.1	Size of the rich clubs in the two connectomes based on how significant the rich-club coefficient is.	18

Chapter 1

Introduction

1.1 Integration and Segregation

How cognition achieved is one of the long lasting questions in the area of neuroscience and network neuroscience. One area of focus has been the occurrence of integration and segregation in a network and how that relates to cognition. With the rise of popularity of network analysis tools and functional brain imaging data, there has been an interest in understanding how both the segregation and the integration of brain networks underlie successful cognition [1].

Functional segregation refers to neuronal processing carried out among functionally related regions arranged within modules [6]. In networks such modules correspond to communities defined by high density of connectivity among members of the same community and low density of connections between members of different communities [6]. It is suggested that during highly focused and specialized tasks, like a delicate motor skill, integration helps better performances [1, 8]. Many different criteria have been suggested to study segregation such as: within module degree [5], modularity [1, 6], system segregation [1], local efficiency [1] and the number of provincial hubs [1]. All of these criteria, try to measure how strongly connected local communities are; which reflects the definition of segregation.

Functional integration on the other hand, can be viewed from at least two different perspectives, one based on the efficiency of global communication (i.e. how fast nodes can propagate data), and another on the ability of the network to integrate distributed information [6]. While the former has a more tangible definition, the latter could be hard to define especially in a static network. However, there has

been efforts to measure integration in networks: participation coefficient [5], hubs [6], rich-club [6], global efficiency [1] and number of connector hubs [1].

Many studies use resting-state fmri, task-based fmri or both at the same time to assess the role of integration and segregation in cognition. [8] used resting state fmri in a large sample of healthy young adults to calculate their networks' segregation and integration. They also measured general and domain-specific cognitive phenotypes from various tasks and were able to show that stronger integration is associated with better general cognitive ability, stronger segregation contributes to crystalized intelligence and processing speed, and an individual's tendency towards balance supports better memory. [1] used both resting-state and task-based fmri for their research. They measured the segregation and integration of participants during a sequence tapping task, which probes motor execution and an n-back task, which is thought to probe working memory and requires coordination across multiple networks. They found that local, within-network communication is critical for motor execution, whereas integrative, between-network communication is critical for working memory. These results show that the human brain has the functional capability of adapting to its needs by changing its functional connectivity. These studies highlight how understanding of both integration and segregation is crucial to the understanding of cognition in networks.

1.2 *Drosophila Melanogaster*

Integration and segregation have mostly been the focus of studies that use functional connectomes. This could be due to the abundance of such cohorts and also lack of enough structural connectomes. However, with the advances in microscopy and high performance segmentation of microscopic data, during the last few years, researchers have been able to discover many structural connectomes, such as drosophila [9, 2], mouse visual system [7], zebra fish [3] and etc. This highlights the opportunity of exploring what information can integration and segregation give us in the context of development.

For this study, we chose drosophila larvae, and drosophila adult brain connectomes to investigate the role of integration and segregation in its development.

Chapter 2

Data Description

2.1 Larvae

This connectome is showing the structure of the 1st instar larvae of female *Drosophila*. You can find it as a supplementary material of [9]. It comes in the form of an adjacency matrix whose rows represent different neurons and entries show the number of synapses between neurons. This matrix is not symmetric and the edges are directional. Along with the adjacency matrix, extra data regarding the neurons are also available, such as cell types, additional annotations and cluster determined by a 7 level hierarchical clustering. It has 3,016 neurons and 548,000 synaptic sites which translate to roughly 100,000 links [2.1](#).

2.2 Adult

This connectome is one of many connectomes published from [FlyWire](#) and it is published under the name of Female Adult Fly Brain (FAFB). In this project, we are using version 783. Similar to the larvae, this is also a female *Drosophila*. This one comes in the form of an edgelist that also contains extra information, namely, neuropil, *syn_count* and *nt_type*. Neuropil refers to a highly dense area of the fly brain that is rich with synapses and neuronal processes. Each edge belongs to one of the 78 distinct neuropils in *Drosophila*. *syn_count* shows the weight of a certain link and refers to the number of synapses between a pair of pre- and post-synaptic neurons. Finally, *nt_type*, is the neurotransmitter at work at that edge/synapse which is classified using machine learning algorithms.

The predicted neurotransmitters are: GABA, glutamate, acetylcholine, serotonin, dopamine and octopamine, which can be used in as an edge class in future studies. This connectome has 139,255 neurons and 50,666,648 synaptic sites which translate to roughly 3,732,460 links [2.1](#).

TABLE 2.1: Summary of Datasets

Dataset	File Type	Nodes	Edges
Larvae	Adjacency Matrix	3,016	100,000
Drosophila	Edge List	139,255	3,732,460

Chapter 3

Basic Stats

In this section, we present the degree distributions and the basic statistics, comprised of average shortest path length and the clustering coefficient.

3.1 Degree Distribution

The following figures show the degree distributions of both the larvae and the adult *Drosophila*. As evident, the larvae has a short tail and it is not a heavy-tail distribution 3.1. This tells us that there are few hubs as there are few nodes with much higher degree than the rest of the nodes. However, the distribution for the adult is following a heavy-tail distribution 3.2. As you can see, most neurons have less than 1000 degree but there are nodes with more than 7000 degree. Those are clear hubs of the network.

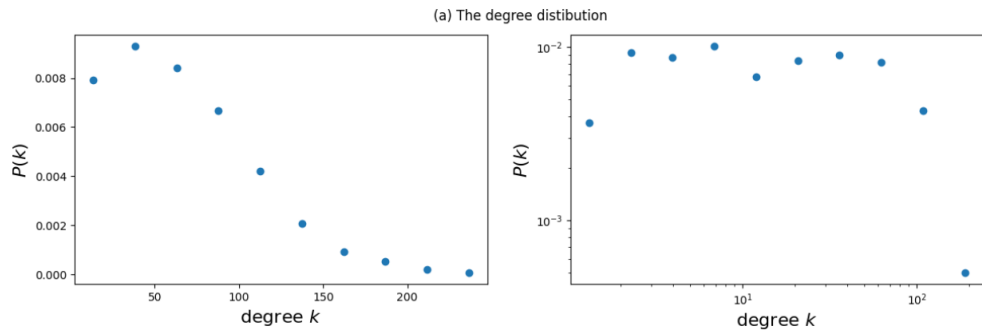


FIGURE 3.1: Larvae degree distribution. Axes are in log scale.

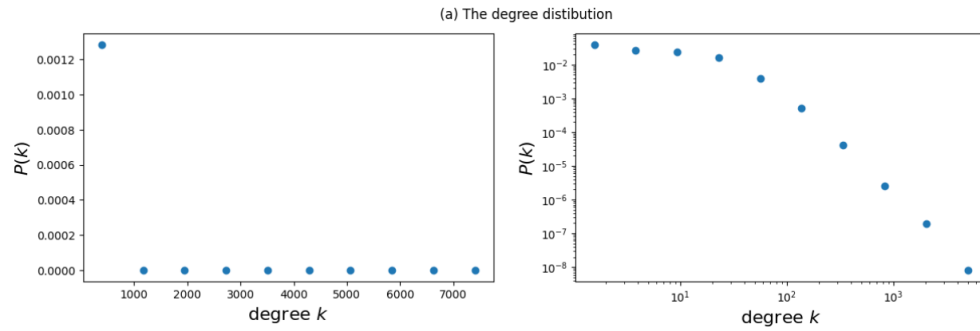


FIGURE 3.2: Adult degree distribution. Axes are in log scale.

3.2 Stats

In this section we calculated the Average Shortest Path Length (ASPL) and the Clustering Coefficient (CC) for both the larvae and the adult. To have an understanding, we also created null models using degree preserving double edge swaps (dp-des). 3.1 shows the stats for all four sets of networks.

TABLE 3.1: Basic Statistics

Dataset	ASPL	ASPL (null)	CC	CC (null)
Larvae	0.450	0.361	0.00453	0.00271
Drosophila	0.293	0.201	0.00205	0.00012

Chapter 4

Methods

4.1 Consensus Clustering

There are many different community detection algorithms, and perhaps, using each one of them we will get a different result for the same problem. That could be due to random decisions made along the way and/or just the fact that they might try to optimize different modularity functions. But if we were to decide to choose one of those partitions, how could we decide which one to choose?

Consensus clustering is a well known technique used to search for a so-called median (or consensus) partition, i.e. the partition that is most similar, on average, to all the input partitions. Using the consensus clustering leads to more accurate partitions and more stable results compared to the partitions delivered by using the methods alone.

We implemented the algorithm following [4.1](#). According to [\[4\]](#), although the algorithm continues until the consensus matrix converges, for most applications only one iteration is enough. So, I first created 100 partitions using the louvain method. However, one issue with the consensus matrix is that it gets dense very fast and that causes memory management difficulties especially for the adult connectome. To reduce the memory footprint, we decided to reduce the number of partitions to 10 so that the density of the consensus matrix stays relatively low and manageable. We then derive a binary adjacency matrix from each of these matrices whose entries determine whether two nodes are in the same community or not. We then sum these adjacency matrices up to get the consensus matrix and then filter the edges with weight less than 7 ($\tau = 0.7$). In order to avoid having a disconnected

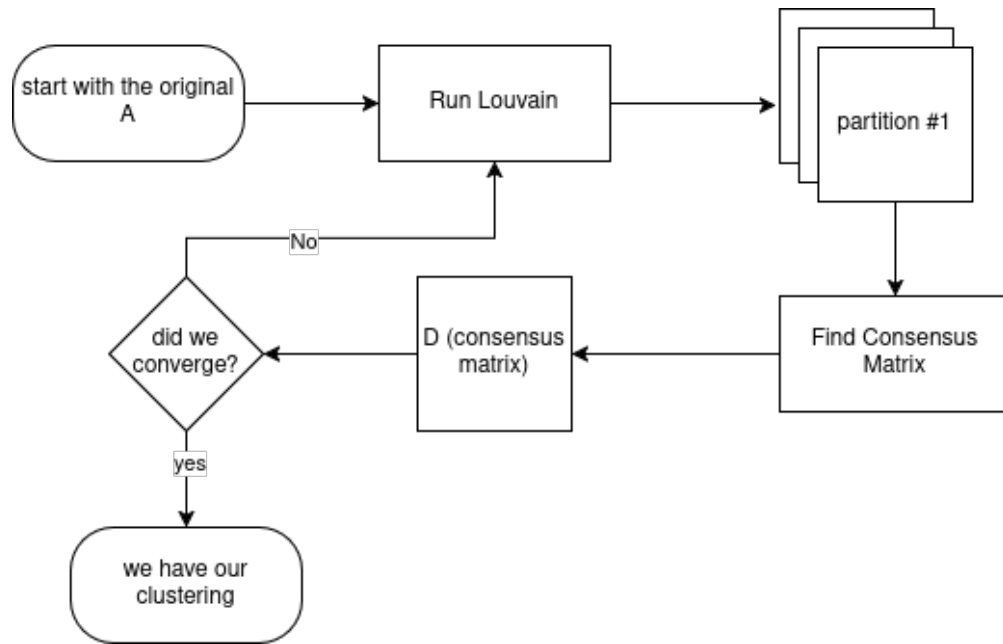


FIGURE 4.1: Diagram explaining the flow of the consensus clustering algorithm.

matrix, if after the thresholding there exists a node that does not have any edges, we retrieve its edge with the highest weight. After finalizing our consensus matrix, we run the louvain algorithm again and these will be our final communities.

Using this method, we created 229 communities for the adult connectome; however, only 8 of these communities had more than 1000 nodes, which shows us 8 large communities of nodes which are highly interconnected and might represent groups of highly specialized neurons which are responsible for different tasks. Surprisingly, this method led to 9 communities for the larvae, 8 of which had more than 100 nodes.

4.2 Null Models

In this section, we first use the derived communities to partition the edges into inter-modular and intra-modular edges. Then, perturbing only one partition at a time, we create 3 sets of null models, aiming to understand the role of various types of edges in the segregation and integration of both larvae and adult *Drosophila*.

We partitioned edges of the connectome into intra-modular, that connect nodes within a module, and inter-modular, that connect nodes from module v to module w , $v \neq w$.

Using this partitioning, we created three sets of null models:

1. Rewire all edges at the same time.
2. Rewire only the inter-modular connections.
3. Rewire only the intra-modular connections.

For rewiring we chose $10 * \text{number_of_edges}$ as our swap count to make sure the structure is randomized without damaging the degree distribution.

To create the first set of null models, we took the entire connectomes and using the library `snap` and method `GenRewire`, we were able to create the null models. Things were just a little more complicated for the other two sets of neurons. We first derived all subgraphs Gm_n for n in $n_modules$. Furthermore, we derived $Gvoid = Gorig - \bigcup_i^{n_modules} Gm_n$. Gm_n contain only the edges that are inside a module and $Gvoid$, is all of the rest of the edges which are the ones that connect the modules together. To create the second set of null models, the ones with rewired inter-modular connections, we left Gm_n intact and rewired only $Gvoid$ and then composed the main null model from $Gvoid$ and all Gm_n . Finally, to create the third set of null models, we perturbed all Gm_n , left $Gvoid$ untouched, and in the end, combined all of them to get the null model. As a result, for each set, we created 100 null models so a total of 300 null models for both larvae and adult connectomes.

4.3 Measures

We calculated the below measures for all of the original networks and the null models:

1. Basic: Clustering Coefficient, Global Efficiency and Local Efficiency
2. Centrality: PageRank, Betweenness, Eigenvector, Closeness
3. Rich-Club Coefficients

In order to do the process, we only used `snap` for the adult connectome and its null models. However, for the larvae, we used `bct` for the basic measures and the rich-club coefficients and used `networkx` only to calculate the centrality measures.

However, to calculate the betweenness centrality and the closeness centrality, which had to do with finding the shortest path between all pairs of nodes, we used a sampling approach instead and chose 1000 random nodes instead and calculated the shortest path among those and the rest of the connectome.

4.4 Parallelization

In order to manage the computations for all 602 networks, We decided to design each procedure so that it takes the address of an edgelist, belonging to one of the networks, does the proper calculation, saves the result as a row in a csv or in a separate file in a specified directory and finishes. Finally, use array jobs to run such script for all 602 networks and gather the final results.

Chapter 5

Visualization

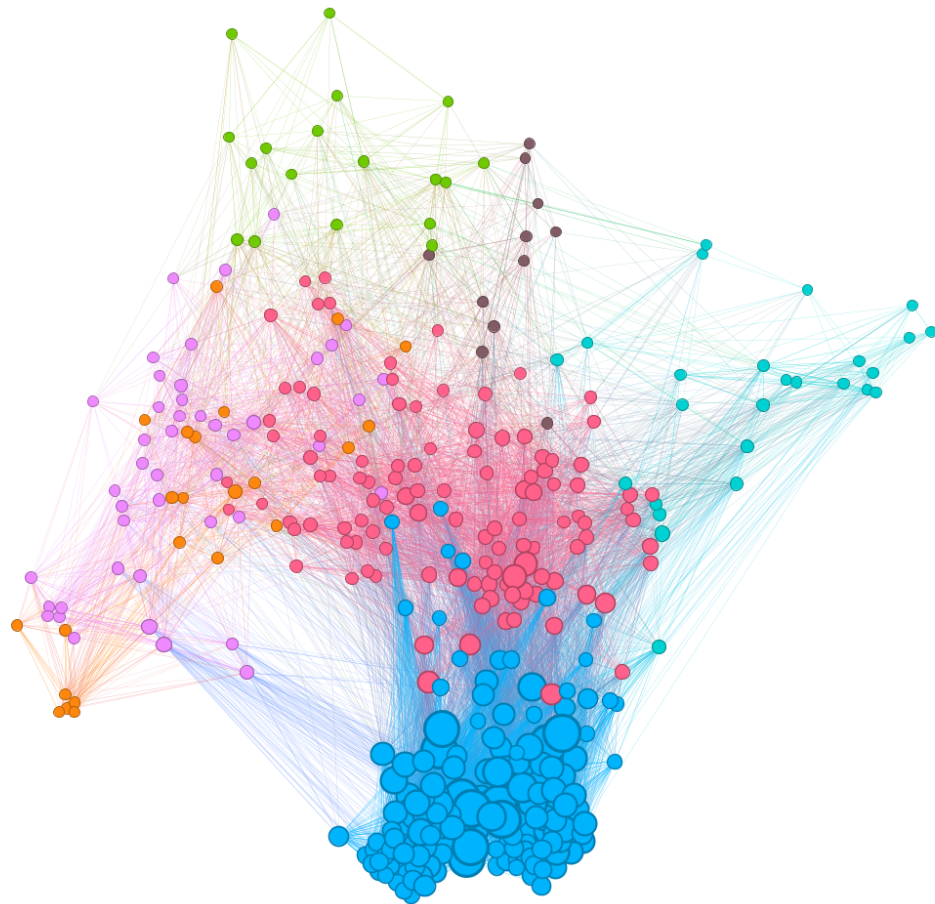


FIGURE 5.1: Gephi visualization for the larvae connectome. Only the top 13.5% of the nodes with the highest degree are visible. Colors show the consensus communities.



FIGURE 5.2: Gephi visualization for the adult connectome. Only the top 0.5% of the nodes with the highest degree are visible. Colors show the consensus communities.

Chapter 6

Results

In this chapter we discuss the results from computation of clustering coefficient and local efficiency, global efficiency, centrality measures and rich-clubs.

6.1 Clustering Coefficient and Local Efficiency

In general, randomizing a network destroys local communities and clusters. That explains why we see on average less clustering in the null models. You can see the clustering coefficient of both connectomes in comparison with their null models plotted in 6.1 and 6.2. Also, We did a scatter plot for local efficiency as the x-axis and the clustering coefficient as the y-axis to see the correlation between the two 6.3.

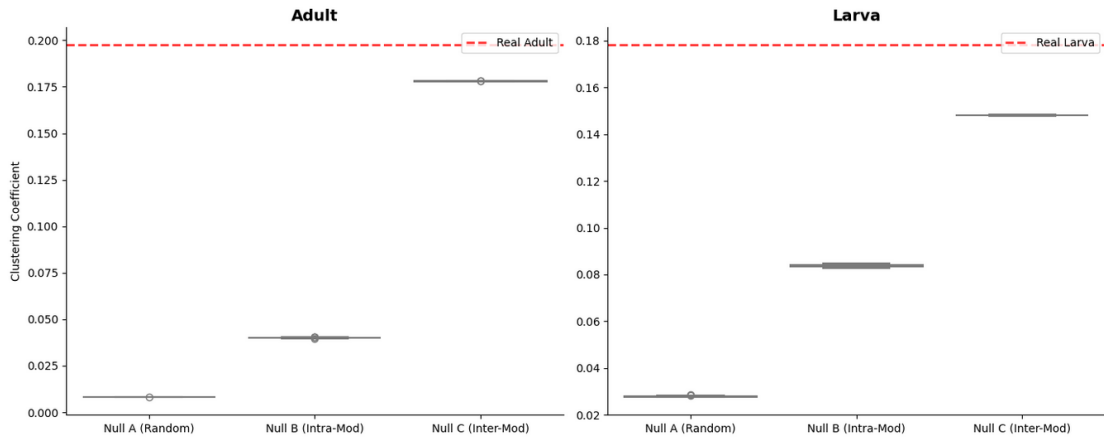


FIGURE 6.1: Box plot comparing the original average clustering coefficient of both connectomes with their null models.

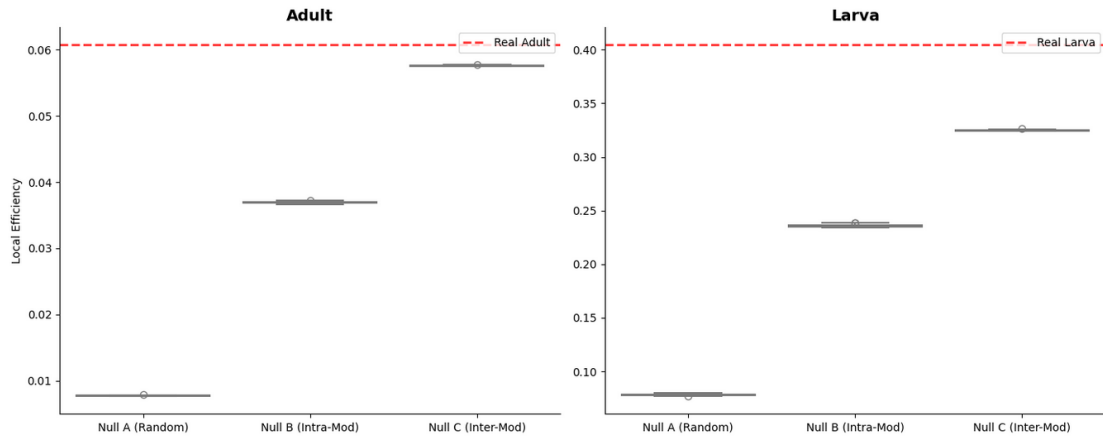


FIGURE 6.2: Box plot comparing the original average local efficiency of both connectomes with their null models.

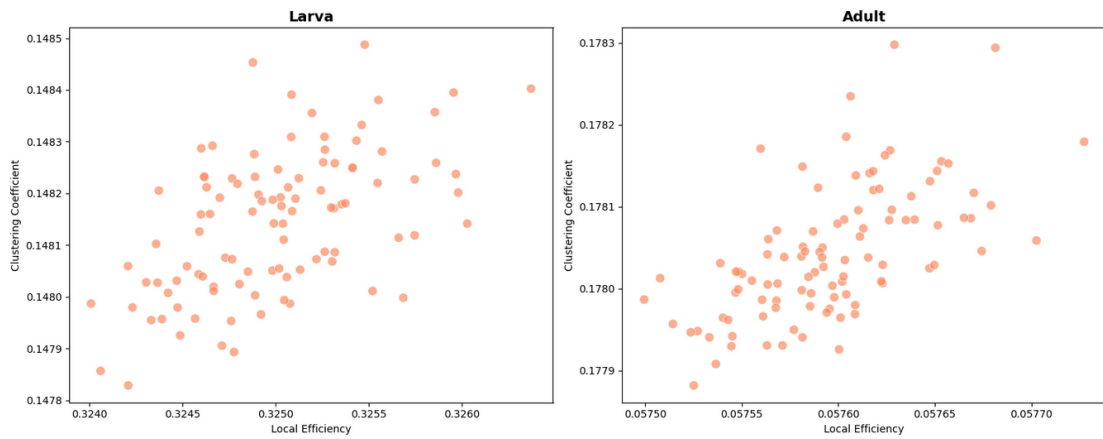


FIGURE 6.3: Scatter plot showing the correlation between the local efficiency and the clustering coefficient in both connectomes.

6.2 Global Efficiency

With randomly rewiring, we add a lot of random shortcuts that in general, reduce the average shortest path of the network and increase the efficiency.

6.3 Centrality Measures

6.3.1 Betweenness Centrality

In 6.5, We can see in average, randomization of any kind is decreasing the betweenness centrality on average. However in the case of max betweenness, it is not the same 6.6. in the case of larvae, randomizing of any kind reduces the maximum

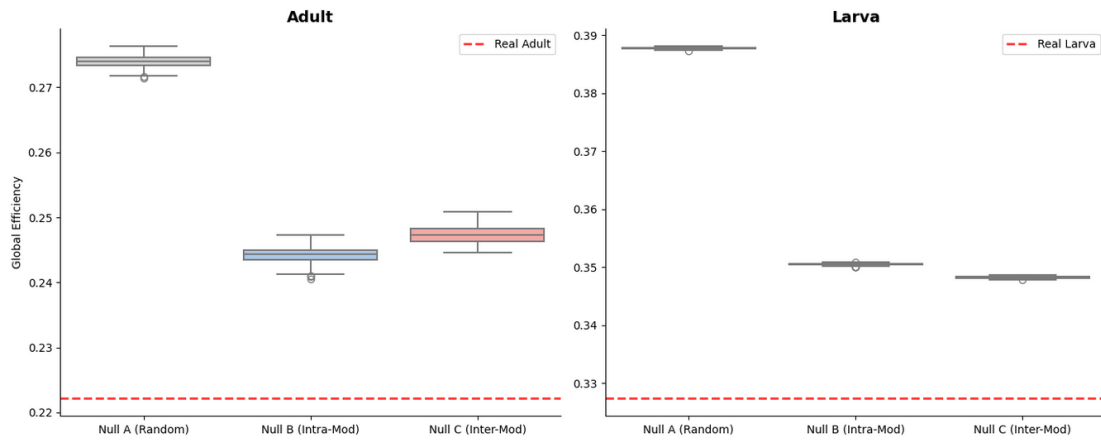


FIGURE 6.4: Box plot comparing the original global efficiency of both connectomes with their null models.

betweenness as well. but in adult we see an interesting pattern. randomizing the inter-modular connections reduced the maximum betweenness and randomizing the intra-modular increased the maximum betweenness. I use an example to explain why I think that happens.

Imagine London and its famous bridges which are highly central. Now, if we randomly rewire its bridges, those bridges won't be as popular as before because they are now making people's travel to work even longer; That is why in Null B, we see the maximum betweenness is decreased. However, if instead, we perturb the local community roads, lets say make more efficient roads that reduces the local traffic (so the efficiency of the local travel is increased), now more people are willing to use the famous bridges to go to work. The reason for that is that, before the local enhancement, they might have preferred to use a less known bridge because the roads leading to that bridge weren't as crowded, but now since there's no traffic even in the roads leading into the famous bridges, they will prefer to choose the famous bridge over the less known bridges. That is why I believe we see the increase in the maximum betweenness in the intra-modular perturbed null models of adult connectome. The reason why that doesn't happen in the larvae is that, perhaps the larvae isn't as segregated as the adult brain and peoples houses and work places are likely to be in the same area and hence the reduction in the max betweenness centrality as well as the mean betweenness centrality.

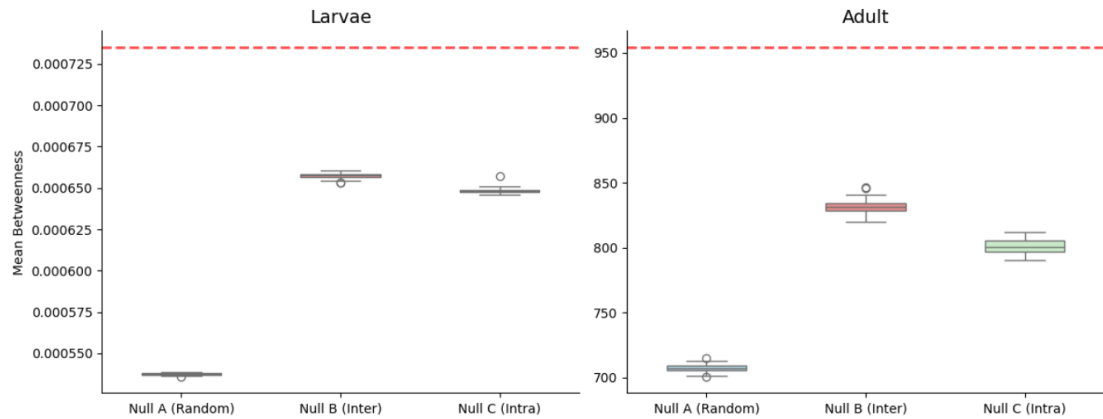


FIGURE 6.5: Mean betweenness of nodes in a network. The box plot is comparing both connectomes with their null models.

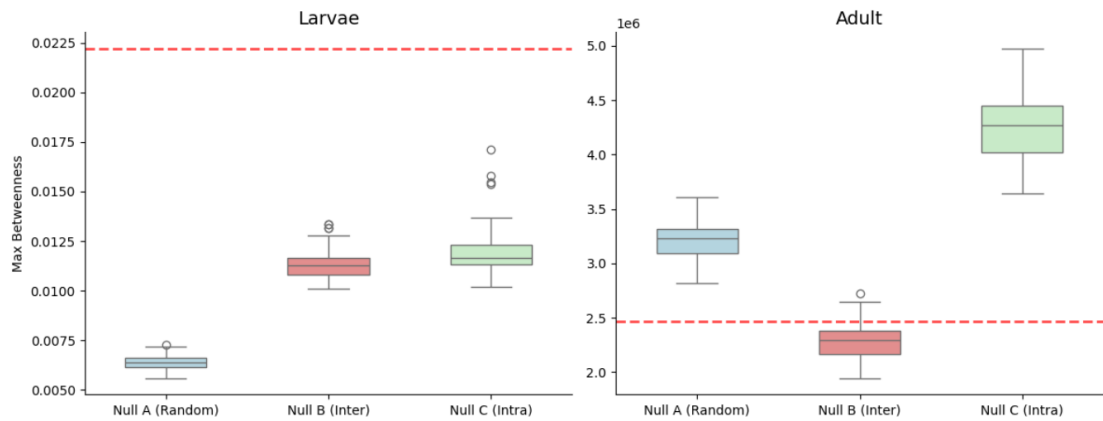


FIGURE 6.6: Maximum betweenness of nodes in a network. The box plot is comparing both connectomes with their null models.

6.3.2 Eigenvector Centrality

When we plot the mean eigenvector centrality for both connectomes compared to their null models (6.7), we notice a difference between how randomizing inter-modular connections have a different effect on the centrality compared to randomizing intra-modular connections.

Lets say the eigenvector centrality is a sign for assortativity. so, an increase in the eigenvector centrality after randomization means that our original network was dis-assortative in the first place and by randomizing its connections, we made it more distributed and assortative. We hypothesize that, this is a result of segregation as there are many provincial hubs in any local community and they are connected not to each other, but to the other local nodes to abilitate local specificity and higher cognition power.

Similarly, when we randomize the inter-modular connections, we see the eigenvector centrality/ assortativity decreases. this means that, now our original network has a highly assortative structure and we are destroying it by randomization and reducing the assortativity. We can relate that to the connection of the connector hubs, usually highly central, to each other, which happens in a highly integrated network and forms rich clubs.

In the plot for the larvae however, we don't see those differences after only inter-/intra-modular randomizations. This could be because of the fact that, the segregation and integration haven't shaped yet in that network.

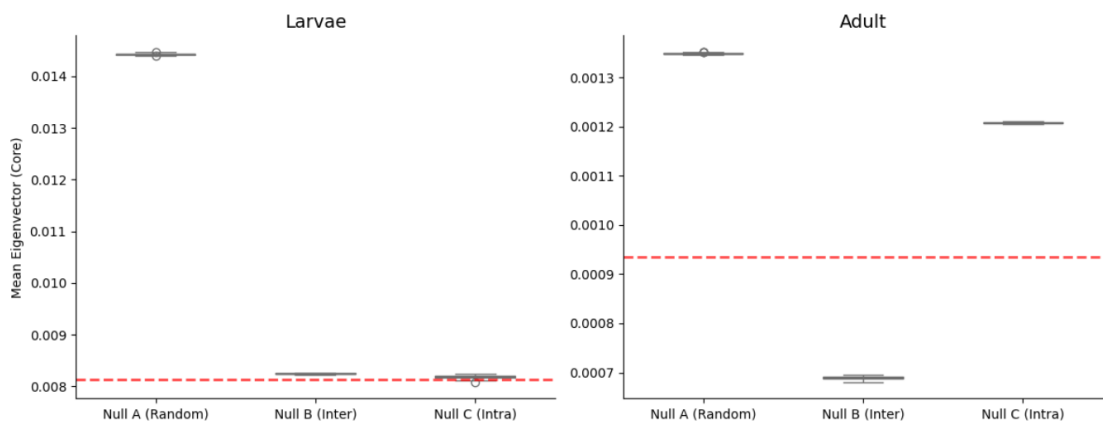


FIGURE 6.7: Mean betweenness of nodes in a network. The box plot is comparing both connectomes with their null models.

6.3.3 Closeness Centrality

Closeness centrality shows similar pattern across all null models and connectomes (6.8). In all cases, randomization helps building shortcuts and that allows for shorter path lengths and higher efficiency and closeness.

6.4 Rich-Club

Here we see the rich club plot, as we can see, both networks show existence of the rich club (6.9). The larvae has two separate rich clubs, the first one spanning from $k=0$ to $k=195$. The same goes for the adult *Drosophila* as well. There is a rich club from $k=0$ to $k=145$ in this connectome. This shows signs of integration in both connectomes as rich clubs are necessary for the integration of information in complex systems. The adult figure is zoomed on $k \leq 1000$.

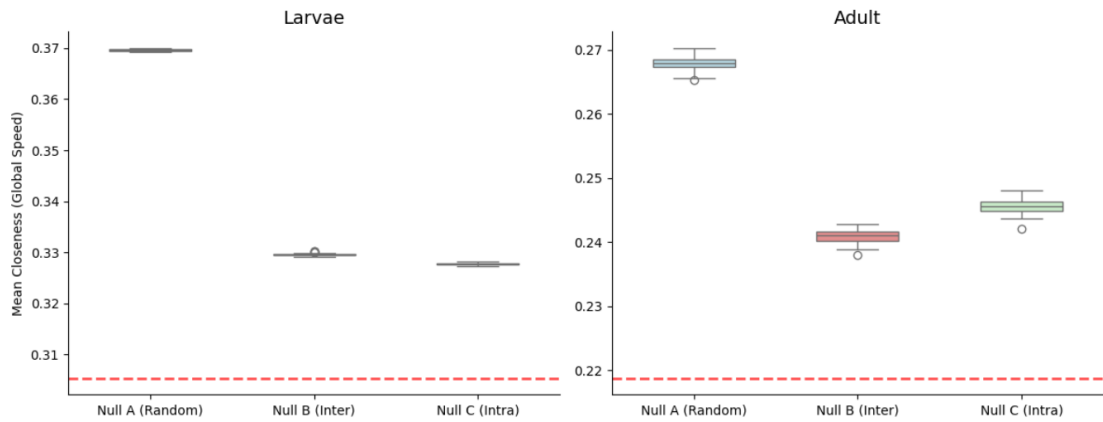


FIGURE 6.8: Mean closeness of nodes in a network. The box plot is comparing both connectomes with their null models.

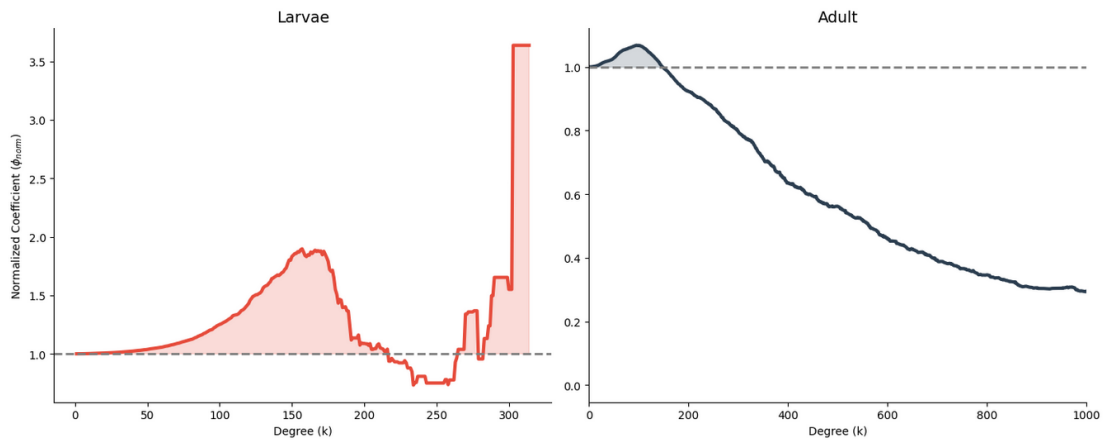


FIGURE 6.9: Mean closeness of nodes in a network. The box plot is comparing both connectomes with their null models.

TABLE 6.1: Size of the rich clubs in the two connectomes based on how significant the rich-club coefficient is.

Connectome	$k(\text{mean} + k * \text{std})$	rich_club_range	neurons_involved
Larvae	0	(0, 215)	0.98442
Larvae	1	(0, 203)	0.97900
Larvae	2	(0, 195)	0.97527
Larvae	3	(0, 195)	0.97527
Adult	0	(0, 146)	0.93887
Adult	1	(0, 145)	0.93829
Adult	2	(0, 145)	0.93829
Adult	3	(0, 145)	0.93829

Chapter 7

Conclusions

In this project, we aimed to find clues and signs for the integration and segregation in *Drosophila* connectomes, during their 1st instar larvae stage and fully developed brain. Previous studies on human connectomes worked on several ways to recognize integration and segregation on fmri data. However, because of availability issues, these studies never focused on structural connectomes. Therefore, in this project, we were dealing with connectomes in different scales. Larvae *Drosophila* has 3,000 nodes and 100,000 links while the Adult *Drosophila* has 140,000 nodes and 3,732,460 links. This causes computational issues as well as methodological ones. On the computational side, the connectivity of the adult brain, while being very sparse, it is very memory heavy. Also many algorithms take too long to run on the adult brain and most often they crash before finish running due to running out of memory. On the methodological side, we need to be content with only the easiest measures to compute and we cannot use more sophisticated methods, both due to the mentioned computational issues and because there's no way to validate their results. We therefore used different libraries to run the same algorithms, we used SNAP to handle the adult connectome whereas we used networkx and BCT to run the processing for the larvae.

We decided to derive three sets of null models as our baseline to compare our original networks to. We divided edges into two partitions; One the edges that connect nodes within the same module, and other the edges that connect nodes from different modules. We call these intra-modular edges and inter-modular edges. We use this definition to make null models that only perturb intra-modular edges, that only perturb inter-modular edges, and the most common of all, that perturbs all edges. We used degree preserving rewiring to derive 100 null models for all three sets of null models for each connectome. Using such setup, we had

the ability to not only see the difference between larvae and the adult, but also see where that difference is made, within modules, or, between modules, hoping to see a pattern for early and late stage development

In our results, we saw no significance in many measures that were claimed to be indicative of integration and segregation, such as clustering coefficient, local efficiency and global efficiency. However we saw results that we thought were interesting, looking at the betweenness centrality, eigenvector centrality and rich-clubs. Betweenness centrality suggested that the adult brain is more segregated while it didn't reveal much about the larvae. Eigenvector centrality, suggested that the adult brain is highly segregated and integrated, while suggesting the opposite for the larvae. Furthermore, we looked at the rich-clubs as well. *Drosophila* both in the larval stage and the adult stage, shows strong signs of existence of rich-clubs. In both connectomes, more than 90% of the neurons were recognized as part of the rich-club, and rich-clubs are clear signs of integration in networks.

While these results are not definite, this shows potential in using similar approaches in studying structural connectomes to find out more about the changes a connectome undergo during development.

Overall, this project allowed me to explore new concepts such as integration and segregation, work with connectomes of different sizes, one being an extremely big connectome which makes us face serious computational issues. Also, I had the chance to try consensus clustering for the first time and see the usefulness firsthand. Furthermore, I had the chance to try a series of null models that focus on different types of edges within the same connectome, which was a novel approach and I believe has more potential to be experimented. While I wasn't able to address the research questions of this project, I see the outcomes of this project as very positive.

Bibliography

- [1] Cohen, J. R. and D’Esposito, M. (2016). The segregation and integration of distinct brain networks and their relationship to cognition. *Journal of Neuroscience*, 36(48):12083–12094.
- [2] Dorkenwald, S., Matsliah, A., Sterling, A. R., Schlegel, P., Yu, S.-C., McKellar, C. E., Lin, A., Costa, M., Eichler, K., Yin, Y., et al. (2024). Neuronal wiring diagram of an adult brain. *Nature*, 634(8032):124–138.
- [3] Hildebrand, D. G. C., Cicconet, M., Torres, R. M., Choi, W., Quan, T. M., Moon, J., Wetzel, A. W., Scott Champion, A., Graham, B. J., Randlett, O., et al. (2017). Whole-brain serial-section electron microscopy in larval zebrafish. *Nature*, 545(7654):345–349.
- [4] Lancichinetti, A. and Fortunato, S. (2012). Consensus clustering in complex networks. *Scientific reports*, 2(1):336.
- [5] Michael, C., Taxali, A., Angstadt, M., Kardan, O., Weigard, A., Molloy, M. F., McCurry, K. L., Hyde, L. W., Heitzeg, M. M., and Sripada, C. (2024). Socioeconomic resources in youth are linked to divergent patterns of network integration/segregation across the brain’s transmodal axis. *PNAS nexus*, 3(9):pgae412.
- [6] Sporns, O. (2013). Network attributes for segregation and integration in the human brain. *Current opinion in neurobiology*, 23(2):162–171.
- [7] The MICrONS Consortium (2025). Functional connectomics spanning multiple areas of mouse visual cortex. *Nature*, 640(8058):435–447.
- [8] Wang, R., Liu, M., Cheng, X., Wu, Y., Hildebrandt, A., and Zhou, C. (2021). Segregation, integration, and balance of large-scale resting brain networks configure different cognitive abilities. *Proceedings of the National Academy of Sciences*, 118(23):e2022288118.

-
- [9] Winding, M., Pedigo, B. D., Barnes, C. L., Patsolic, H. G., Park, Y., Kazimiers, T., Fushiki, A., Andrade, I. V., Khandelwal, A., Valdes-Aleman, J., et al. (2023). The connectome of an insect brain. *Science*, 379(6636):eadd9330.

Code Availability

For the code that was used to work on this project, please refer to [this](#) github repository.